

Vectors

GCSE Maths · Practice Questions

20 questions · 62 marks total

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Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1 [1 mark]

Write the translation '3 units right and 2 units down' as a column vector.

.....

2 [2 marks]

Work out $\begin{pmatrix} 2 \\ 5 \end{pmatrix} + \begin{pmatrix} 3 \\ -1 \end{pmatrix}$

.....

3 [2 marks]

Work out $\begin{pmatrix} 6 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 5 \end{pmatrix}$

.....

4 [2 marks]

Work out $3 \begin{pmatrix} 2 \\ -4 \end{pmatrix}$

.....

5 [3 marks]

$\mathbf{a} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$

Work out $\mathbf{a} + 2\mathbf{b}$

.....

6

[3 marks]

$$\mathbf{a} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \text{ and } \mathbf{b} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$$

Work out $2\mathbf{a} - \mathbf{b}$

7

[3 marks]

A is the point (1, 2). A is translated by $\begin{pmatrix} 4 \\ -3 \end{pmatrix}$ to give B.

(a) Write down the coordinates of B.

(b) Write down the vector that would translate B back to A.

8

[3 marks]

Work out the magnitude (length) of

(a) $\begin{pmatrix} 6 \\ 8 \end{pmatrix}$ (b) $\begin{pmatrix} 5 \\ 12 \end{pmatrix}$

9

[4 marks]

The vector from O to A is $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ and the vector from O to B is $\begin{pmatrix} 5 \\ 7 \end{pmatrix}$

(a) Work out the vector from A to B.

(b) Work out the length of AB.

10

[4 marks]

(a) Show that $\begin{pmatrix} 4 \\ 6 \end{pmatrix}$ and $\begin{pmatrix} 6 \\ 9 \end{pmatrix}$ are parallel.

(b) Write $\begin{pmatrix} 10 \\ 15 \end{pmatrix}$ as a multiple of $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$

1 [2 marks]

$$\mathbf{a} = \begin{pmatrix} 4 \\ -1 \end{pmatrix} \text{ and } \mathbf{b} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$$

Work out $3\mathbf{a} + 2\mathbf{b}$

.....

2 [2 marks]

Work out the magnitude of $\begin{pmatrix} -7 \\ 24 \end{pmatrix}$

.....

3 [3 marks]

In triangle OAB, the vector from O to A is $\mathbf{a}$ and the vector from O to B is $\mathbf{b}$.

Write the vector from A to B in terms of $\mathbf{a}$ and $\mathbf{b}$.

.....

4 [3 marks]

In triangle OAB, $OA = \mathbf{a}$ and $OB = \mathbf{b}$. M is the midpoint of AB.

Write the vector from O to M in terms of $\mathbf{a}$ and $\mathbf{b}$.

.....

5 [4 marks]

In triangle OAB, $OA = \mathbf{a}$ and $OB = \mathbf{b}$. P lies on AB so that $AP : PB = 1 : 3$

Write the vector from O to P in terms of $\mathbf{a}$ and $\mathbf{b}$.

.....

6 [4 marks]

$OC = 3\mathbf{a} + 2\mathbf{b}$ and $OD = 6\mathbf{a} + 4\mathbf{b}$.

Show that O, C and D lie on the same straight line.

.....

7

[4 marks]

$OA = 2\mathbf{a} + \mathbf{b}$, $OB = 4\mathbf{a} + 3\mathbf{b}$ and $OC = 7\mathbf{a} + 6\mathbf{b}$.

Show that A, B and C lie on the same straight line.

.....

8

[4 marks]

OACB is a parallelogram with $OA = \mathbf{a}$ and $OB = \mathbf{b}$.

M is the midpoint of AC.

Write the vector from O to M in terms of $\mathbf{a}$ and $\mathbf{b}$.

.....

9

[4 marks]

In triangle OAB, $OA = \mathbf{a}$ and $OB = \mathbf{b}$. P lies on AB with $AP : PB = 2 : 3$

Show that $OP = \frac{1}{5}(3\mathbf{a} + 2\mathbf{b})$

.....

10

[5 marks]

OABC is a parallelogram with $OA = \mathbf{a}$ and $OC = \mathbf{c}$.

M is the midpoint of AB and N is the midpoint of BC.

(a) Write OM in terms of $\mathbf{a}$ and $\mathbf{c}$.

(b) Write ON in terms of $\mathbf{a}$ and $\mathbf{c}$.

(c) Show that MN is parallel to AC, and state the ratio of their lengths.

.....